

TI-84 Quick Guide: One-Variable Statistics

The TI-84 calculator can quickly compute important statistics such as the sample size, sum of the data, sum of squares, mean, and standard deviation.

Example

Suppose the quiz scores of five students are:

4, 6, 8, 10, 12

We will use the TI-84 to find:

- Sample size (n)
- Sum of the data ($\sum x$)
- Sum of squares ($\sum x^2$)
- Mean ($\bar{x}$)
- Sample standard deviation (s)

Step 1: Enter the Data

Press:

STAT

Select:

1: Edit

Enter the data into list L1:

L1
4
6
8
10
12

Step 2: Calculate Statistics

Press:

STAT

Use the right arrow key to move to:

CALC

Select:

1: 1-Var Stats

Type:

L1

(To enter L1, press 2nd and then 1.)

The screen should display:

1-Var Stats L1

Press:

ENTER

Calculator Output

The calculator will display statistics similar to:

$$\bar{x} = 8$$

$$\Sigma x = 40$$

$$\Sigma x^2 = 360$$

$$S_x = 3.1623$$

$$\sigma_x = 2.8284$$

$$n = 5$$

Interpreting the Results

Sample Size

$$n = 5$$

There are five observations in the data set.

Sum of the Data

$$\sum x = 40$$

Since

$$4 + 6 + 8 + 10 + 12 = 40,$$

the calculator reports $\sum x = 40$.

Sum of Squares

$$\sum x^2 = 360$$

The calculator computes

$$\begin{aligned} &4^2 + 6^2 + 8^2 + 10^2 + 12^2 \\ &= 16 + 36 + 64 + 100 + 144 \\ &= 360. \end{aligned}$$

Mean

$$\bar{x} = 8$$

The average quiz score is 8.

Sample Standard Deviation

$$s = S_x = 3.1623$$

Use S_x when the data represent a sample.

Population Standard Deviation

$$\sigma = \sigma_x = 2.8284$$

Use σ_x only when the data represent an entire population.

Important TI-84 Outputs

Quantity	TI-84 Output
Number of observations	n
Sum of the data	Σx
Sum of squares	Σx^2
Mean	$\bar{x}$
Sample standard deviation	S_x
Population standard deviation	σ_x

Practice

The daily number of customers at a coffee shop for six days was

$$12, 15, 18, 20, 25, 30.$$

Enter the data into L1, run 1-Var Stats, and record:

1. n
2. Σx

3. $\sum x^2$

4. $\bar{x}$

5. s

Tip: The command **1-Var Stats** is one of the most useful tools on the TI-84. It provides many of the summary statistics that we will use throughout this course.